Fall 2026 Section: _____ Name: _____

# MIDTERM 2

Read the following directions carefully. DO NOT turn to the next page until the exam has started.

Write your name and section number at the top right of this page:

| Class | Section Number |
|---|---|
| Sahifa 12:05 | 002 |
| Sahifa 1:20 | 003 |

As you complete the exam, write your initials at the top right of each other page.

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When the exam start time is called, you may turn the page and begin your exam.
If you need more room, there is a blank page at the end of the exam, or we can give you some scratch paper.

Some multiple choice questions are “Select ONE" while others are “Select ALL that apply". Pay attention to the question type and only mark one option if it says “Select ONE". Fill in the circles completely.

If you finish early, you can hand your exam to your instructor or TA and leave early.

Otherwise, stop writing and hand your exam to your instructor or TA when the exam stop time is called.

1. The pmf of discrete random variable $X$ is given in the table below.

| $x$ | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| $P(X = x)$ | 0.2 | 0.4 | 0.15 | 0.05 | 0.2 |

$X$ has expected value $\mu = 2.65$.

(a) (3 points) What can be said about the 25th and the 50th percentile? **Select ALL**.

- ◯ The 25th percentile is equal to 1.
- ● The 50th percentile is equal to 2.
- ● The 25th percentile is same as the 50th percentile.
- ◯ The 25th percentile is a number between 1 and 2.
- ◯ The 50th percentile is a number between 1 and 2.
- ◯ Cannot be determined from information given

(b) (1 point) If X was a continuous random variable, would your answer for part (a) be the same? **Select ONE.**

- ◯ Yes
- ● No

(c) (2 points) What is the probability that X is equal to $\mu$? **Briefly explain your answer.**

**Solution:**

$$P(X = \mu) = P(X = 2.65) = 0$$

X can not take the value of 2.65 since it is not in the sample space or distribution.

(d) (1 point) Is it possible to say that X is a binomial random variable? **Give one reason for your decision.**

**Solution:** X cannot be a binomial random variable since the sample space of X doesn't list the possibility of getting no successes in the 5 trials.

2. (a) (5 points) Decide for each of the following random variables if it is binomial or not. **If it is binomial decide what should be the parameters and if it is not binomial mention why.**

   (i) Number of candidates a corporate recruiter interviews until they successfully hire 3 software engineers.

   **Solution:** Not a binomial random variable.

   (ii) Number of policyholders who file a claim in 2026 out of the 25 Aaron manages. Historical data shows that 8% of policyholders file a claim in a given year.

   **Solution:** Binomial. Binom$(25, 0.08)$

   (iii) Number of phone calls that a call center gets in an hour. Assuming that calls occur independently have a constant rate of 25 calls per hour.

   **Solution:** Not a Binomial rv.

   (iv) Number of liver cancer patients, out of a sample of 1,000, correctly identified as having liver cancer by a machine learning model with an accuracy of 90%.

   **Solution:** Binomial. Binom$(1000, 0.9)$

   (v) Number of defective bulbs in a batch of 20 produced in a factory with a 3% defect rate.

   **Solution:** Binomial. Binom$(20, 0.03)$

   (b) (2 points) Out of the variables binomially distributed in (a), which of those can be most accurately approximated by a normal distribution? **Mention the exact normal distribution along with the parameters. You may leave your answer as an expression.**

   **Solution:** The random variable mentioned in (iv) is best approximated as a normal random variable since the variance, $np(1-p) = 90$ is greater than 10.

$$N(1000 * 0.9, \sqrt{1000 * 0.9 * 0.1})$$

3. You have received two offers for a senior engineer position and would like to compare the salaries offered in Boston and Madison.

   Assume that the annual salaries of senior engineers in Boston are normally distributed with a mean of $90,000 and a standard deviation of $5,000. In Madison, assume that annual salaries of senior engineers are normally distributed with a mean of $65,000 and a standard deviation of $10,000.

   (a) (3 points) The annual salary offered in Boston is $100,000, while the offer in Madison is $70,000. Based on the relative standing of each offer within its respective salary distribution, which offer is more favorable? **Justify your answer using z-scores.**

   **Solution:** The z-score is calculated as:

   $$z = \frac{x - \mu}{\sigma}$$

   Boston offer:

   $$z_B = \frac{100000 - 90000}{5000} = 2$$

   Madison offer:

   $$z_M = \frac{70000 - 65000}{10000} = 0.5$$

   The Boston offer is more favorable in terms of relative salary standing because it is 2 standard deviations above the Boston mean, compared with only 0.5 standard deviations above the Madison mean.

   (b) (3 points) Suppose you prefer to work in Madison. What annual salary would you need to negotiate in Madison to have the same percentile ranking within the Madison salary distribution as the $100,000 offer in Boston? Which of the following R code expressions correctly calculates this salary? **Select ONE.**

   - ◯ `qnorm(100000, 90000, 5000) %>% pnorm(65000, 10000)`
   - ● `pnorm(100000, 90000, 5000) %>% qnorm(65000, 10000)`
   - ◯ `pnorm(70000, 65000, 10000) %>% qnorm(90000, 5000)`
   - ◯ `qnorm(70000, 65000, 10000) %>% pnorm(90000, 5000)`

   (c) (2 points) If the average salary of the 10 senior engineers working for the company in Madison is $75,000. What R code should be used to get the percentile of this average salary? **Select ONE**.

   - ◯ `pnorm(75000, 65000, 10000, lower.tail = F)`
   - ◯ `pnorm(75000, 65000, 10000))`
   - ◯ `pnorm(75000, 65000, 10000/sqrt(10), lower.tail = F)`
   - ● `pnorm(75000, 65000, 10000/sqrt(10))`

4. Suppose you are curious about the mean score of this statistics exam. You asked me about it and I didn't tell you what the mean was. The only information i gave you was the standard deviation of the test which was 3 points

   You ultimately wish to create a 90% confidence interval for the mean score. So you collect all 10 of your friends' test scores which gives you an average of 37.5 out of a total of 40.

   (a) (2 points) What will be the point estimate and sampling distribution for your confidence interval. **Use the numbers mentioned above in the distribution.**

   **Solution:**

   P.E: $\overline{x} = 37.5$

   Sampling distribution: $N(\mu, \frac{3}{\sqrt{10}})$

   (b) (2 points) Fill in the blanks to get an R code for the critical value.

   cv = ___qnorm___(___0.95___)

   (c) (2 points) Choose the correct critical value from the table below and write the expression of the lower endpoint and upper endpoint of the desired 90% confidence interval. **You do NOT need to simplify your answer. You can leave at as an expression.**

| Percentile | Critical value |
|---|---|
| 0.95 | 1.645 |
| 0.975 | 1.96 |
| 0.995 | 2.58 |

   **Solution:**

   Lower Endpoint:

   $$37.5 - 1.645 \times \frac{3}{\sqrt{10}}$$

   Upper Endpoint:

   $$37.5 + 1.645 \times \frac{3}{\sqrt{10}}$$

5. A cable TV provider wants to assess customer satisfaction. If less than 60% of its customers are satisfied with the service, the provider will change its pricing scheme.

   A sample of 85 random customers finds that 46 of them are satisfied with the cable TV service.

   (a) (2 points) Identify null and alternative hypotheses to answer the provider's question. The hypotheses should be in terms of $p$, the true proportion of customers who are satisfied.

   **Solution:** $H_0 : p = 0.6$ and $H_\alpha : p < 0.6$

   (b) (1 point) Considering the hypotheses you constructed in part (a), what would be your choice for point estimate that can be used in the test? Briefly explain your choice.

   **Solution:** P.E: $\hat{p}$ = sample proportion or $X$ = Number of satisfied customer.

   (c) (2 points) Assuming that the null distribution of the test statistic ($ts$) is normal. What should be the R code that should be used to find the p-value needed for the test? **Select ONE**.

   - ◯ `pnorm(TS, lower.tail = F)`
   - ◯ `2 × pnorm(TS, lower.tail = F)`
   - ● `pnorm(TS)`
   - ◯ `2 × pnorm(TS)`

   (d) (3 points) The p-value of the test is about 0.16. What are true statements or valid conclusions we can make based on this result at the 5% level? **Select ALL that apply**.

   - ● We know that less than 60% of customers in the sample are satisfied.
   - ◯ We know that less than 60% of all customers are satisfied.
   - ● We fail to reject the null hypothesis, $H_0$.
   - ◯ We accept the alternative hypothesis, $H_\alpha$.
   - ◯ There's insufficient evidence to support $H_0$.
   - ◯ There's sufficient evidence to support $H_0$.

6. Below are some statements about Type I & II error, $\alpha$ and $\beta$. Identify which of these statements are wrong. **Select ALL that apply**.

- ◯ A Type I error occurs when we reject a null hypothesis that is actually true.
- ● The significance level $\alpha$ is the complement of $\beta$.
- ● If $\alpha = 0.05$, there is a 5% probability that the null hypothesis is true.
- ◯ The probability of making a type II error is $\beta$.
- ● Decreasing the sample size reduces the probability of making a Type I & II error.
- ◯ If $\alpha = 0.01$, we are willing to accept a 1% chance of making a Type I error.